



HOSPITAL APPOINTMENT SCHEDULING OPTIMIZATION

Abstract

This project presents a Hospital Appointment Scheduling System that automates the process of booking and managing patient appointments. It is designed to reduce waiting time, handle emergency cases efficiently, and improve doctor utilization.

The system uses Flask and SQLite to manage doctors, appointments, and a waiting queue. Patients are assigned priorities based on urgency levels such as Emergency, High, Medium, and Low.

To improve scheduling, the system applies Greedy Algorithm, Genetic Algorithm, and Simulated Annealing, which help in fast decision-making, better optimization, and conflict resolution.

Overall, the system provides a smart and efficient way to manage hospital appointments, improving both patient experience and resource management.

Introduction

Hospital appointment scheduling is an important task that affects patient satisfaction and doctor efficiency. Traditional methods are often manual and can lead to long waiting times, poor handling of emergency cases, and uneven workload among doctors.

This project introduces an automated scheduling system that uses optimization techniques to assign appointments based on patient priority, doctor availability, and time slots. It aims to make the scheduling process faster, more efficient, and more reliable.

The system also supports priority-based handling of patients and reduces scheduling conflicts. By automating the process, it minimizes human effort and improves overall hospital management.

Problem Statement

Hospital appointment scheduling is often done manually or with basic systems, which leads to long waiting times, poor handling of emergency cases, and inefficient use of doctors' time. It becomes difficult to manage patient priorities, avoid scheduling conflicts, and balance workload among doctors.

There is a need for an intelligent system that can automatically schedule appointments by considering patient urgency, doctor availability, and time constraints. The system should reduce waiting time, improve resource utilization, and provide efficient and fair scheduling for all patients.

Objectives

- ◆ To develop a system for automating hospital appointment scheduling.
- ◆ To reduce patient waiting time and improve service efficiency
- ◆ To prioritize patients based on urgency (Emergency, High, Medium, Low)
- ◆ To optimize doctor utilization and balance workload
- ◆ To minimize scheduling conflicts using optimization algorithms

METHODOLOGY

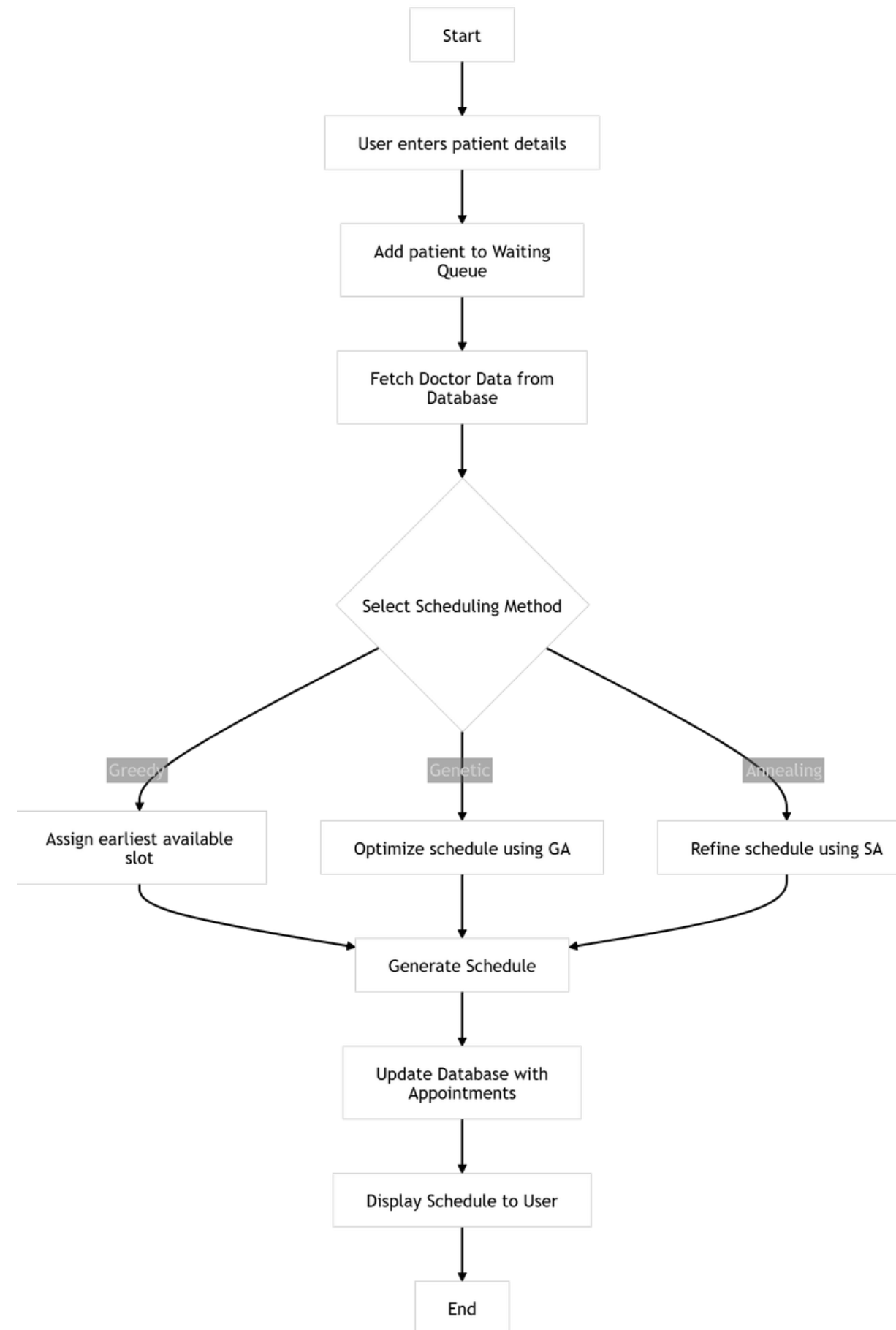
Proposed System / Solution

The proposed system, a Hospital Appointment Scheduling System, is designed as a web-based application that automates the process of booking and managing patient appointments. The system allows users to enter patient details, select priorities, and view available doctor slots through a simple interface. It manages patient data, doctor availability, and appointment records using a structured database.

Once the data is entered, the system processes it through a scheduling engine that assigns appointments based on priority and availability. Optimization techniques such as Greedy Algorithm, Genetic Algorithm, and Simulated Annealing are used to generate efficient and conflict-free schedules. The system also maintains a waiting queue for patients when slots are unavailable.

Finally, the generated schedule is displayed to the user, and the system is evaluated based on reduced waiting time, better resource utilization, and improved scheduling efficiency.

WORKFLOW



Technical Specifications

- Frontend:
HTML, CSS (for designing user interface)
- Backend:
Flask (Python-based web framework)
- Programming Language:
Python
- Database:
SQLite (for storing patient, doctor, and appointment data)
- Algorithms Used:
Greedy Algorithm, Genetic Algorithm, Simulated Annealing
- Development Tools:
VS Code / Any Code Editor

Implementation

Hospital Appointment Booking Optimization

👤 Patient Appointment Booking

Choose Doctor (Type Name)

Start typing doctor name...

Preferred Time

-- : --

Urgency

Low

Book Appointment

📅 Booked Appointments

Doctor	Time	Urgency
Dr Rahul Patil (Cardiologist)	10:16	Low
Dr Rahul Patil (Cardiologist)	10:29	Low
Dr Rahul Patil (Cardiologist)	10:00AM	Medium
Dr Rahul Patil (Cardiologist)	10:44	Medium
Dr Rahul Patil (Cardiologist)	10:44	Medium
Dr Sneha Joshi (Neurologist)	12:32	Medium

Search

hp

99+

ENG IN

30°C Mostly clear

21:45 22-03-2026

Implementation

Booked Appointments

Doctor	Time	Urgency
Dr Rahul Patil (Cardiologist)	10:16	Low
Dr Rahul Patil (Cardiologist)	10:29	Low
Dr Rahul Patil (Cardiologist)	10:00AM	Medium
Dr Rahul Patil (Cardiologist)	10:44	Medium
Dr Rahul Patil (Cardiologist)	10:44	Medium
Dr Sneha Joshi (Neurologist)	12:32	Medium
Dr Rahul Patil (Cardiologist)	10:10	Low
Dr Rahul Patil (Cardiologist)	10:00	Low
Dr Rahul Patil (Cardiologist)	10:02	Low
Dr Sandeep Patil (Orthopedic)	9:00AM	High

Automated Scheduling

Step 1: Add patients to waiting queue, then run automated scheduling.

Add Patient to Queue

Patient Name

Implementation

Hospital Appointment Booking Op X

127.0.0.1:5000

Chat

 Automated Scheduling

Step 1: Add patients to waiting queue, then run automated scheduling.

Add Patient to Queue

Patient Name

Enter patient name

Urgency Level

🔥 Medium

Preferred Time (Optional)

--:--

Add to Queue

 Patients in Queue

Patient Name	Urgency	Preferred Time
No patients in queue		

Run Auto Scheduling

Choose Algorithm:

Search

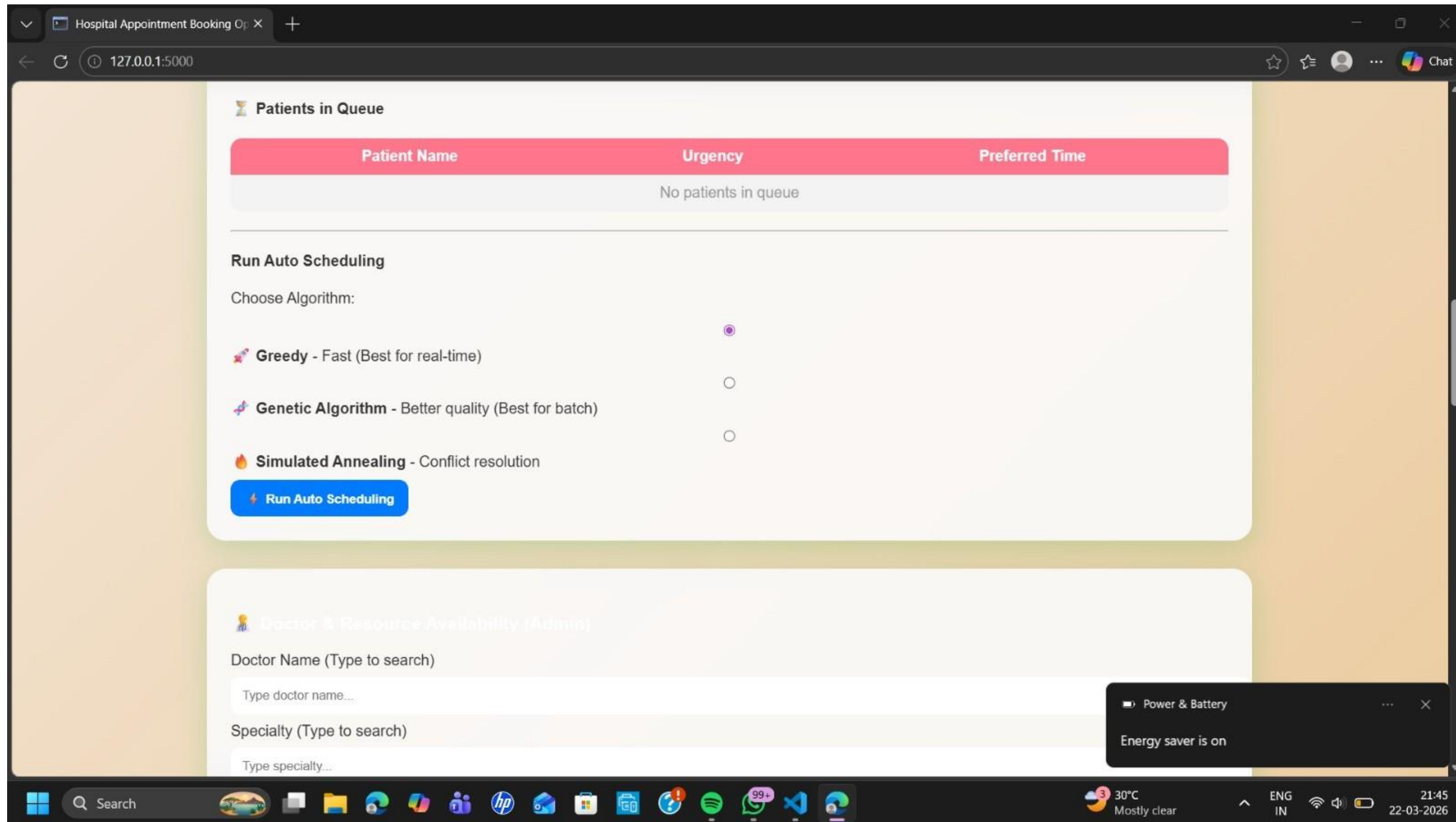


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21:45
22-03-2026

Implementation



Implementation

Specialty (Type to search)

Type specialty...

Available Time Slot

Example: 10AM - 2PM

Add Doctor Availability

Doctor Schedules

Name	Specialty	Available Time
Dr Rahul Patil	Cardiologist	10AM-2PM
Dr Sneha Joshi	Neurologist	11AM-3PM
Dr Sandeep Patil	Orthopedic	9AM-1PM
Dr Neha Kulkarni	Gynecologist	10AM-4PM
Dr Amit Deshmukh	Dermatologist	12PM-5PM
Dr Pooja Shah	Pediatrician	9AM-12PM
Dr Anil Pawar	ENT	11AM-2PM
Dr Ritu Jain	Dentist	10AM-1PM
Dr Kunal Mehta	Psychiatrist	3PM-6PM
Dr Varsha Patil	Physician	9AM-3PM
Dr Nikhil Patil	Urologist	1PM-5PM
Dr Manish More	Surgeon	10AM-2PM

Power & Battery

Energy saver is on

Hot days ahead 30°C

ENG IN

21:45 22-03-2026

Implementation

Hospital Appointment Booking

127.0.0.1:5000

Dr Anil Pawar	ENT	11AM-2PM
Dr Ritu Jain	Dentist	10AM-1PM
Dr Kunal Mehta	Psychiatrist	3PM-6PM
Dr Varsha Patil	Physician	9AM-3PM
Dr Nikhil Patil	Urologist	1PM-5PM
Dr Manish More	Surgeon	10AM-2PM
Dr Kavita Desai	Eye Specialist	11AM-4PM
Dr Rohit Joshi	Cardiologist	9AM-1PM
Dr Aarti Kulkarni	Neurologist	2PM-6PM
Dr Sunil Patil	Orthopedic	10AM-3PM
Dr Meenal Shah	Physician	9AM-12PM
Dr Pravin Pawar	ENT	12PM-4PM
Dr Seema Jain	Gynecologist	10AM-2PM
Dr Akash Mehta	Dermatologist	1PM-5PM

Doctor Workload Dashboard

Load Dashboard

Power & Battery

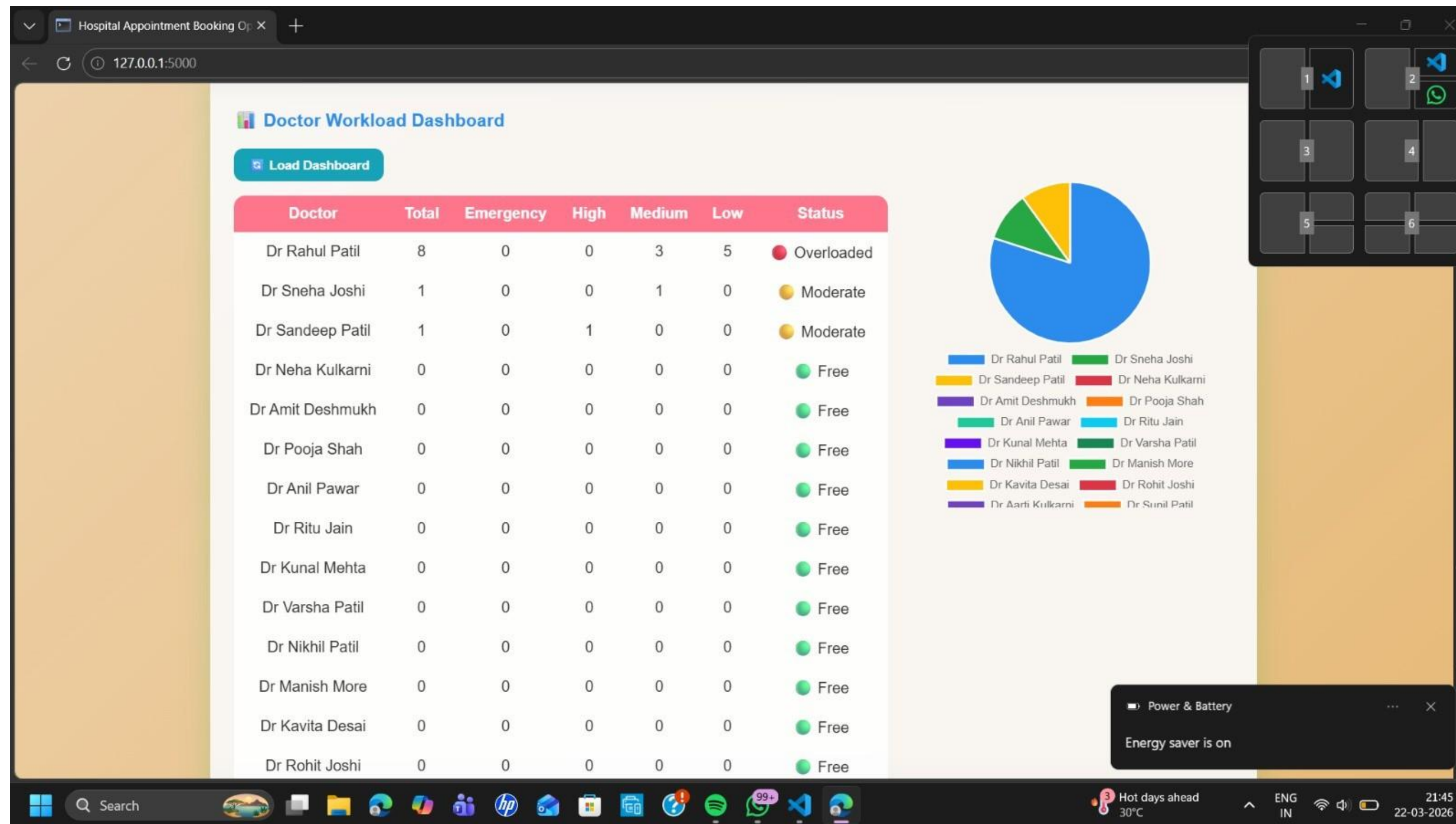
Energy saver is on

Hot days ahead 30°C

ENG IN

21:45 22-03-2026

Implementation



Implementation

Doctor Panel

Select Doctor

Dr Rahul Patil

[View Schedule](#)

Doctor: Dr Rahul Patil

Total Patients: 8

Time	Urgency
10:00	Low
10:00AM	Medium
10:02	Low
10:10	Low
10:16	Low
10:29	Low
10:44	Medium
10:44	Medium

The screenshot shows a web browser window with the title 'Hospital Appointment Booking Op'. The address bar shows '127.0.0.1:5000'. The page content includes a 'Doctor Panel' section with a 'Select Doctor' dropdown menu currently showing 'Dr Rahul Patil'. Below this is a purple 'View Schedule' button. Further down, it says 'Doctor: Dr Rahul Patil' and 'Total Patients: 8'. A table displays the appointment schedule with two columns: 'Time' and 'Urgency'. The table contains eight rows of data. The Windows taskbar is visible at the bottom, showing various application icons and system information like 'Hot days ahead 30°C', 'ENG IN', and the date '22-03-2026'.

Advantages

- Reduces Waiting Time:
Efficient scheduling ensures patients are attended quickly.
- Priority-Based Handling:
Emergency and high-priority patients are given immediate attention.
- Optimized Resource Utilization:
Better allocation of doctors and time slots avoids overload and idle time.
- Minimizes Scheduling Conflicts:
Advanced algorithms help in avoiding double bookings and overlaps.
- Automated System:
Reduces manual effort and human errors in appointment management.
- Improved Patient Experience:
Faster service and organized scheduling enhance satisfaction.

Future Scope:

- **AI-Based Smart Scheduling:**
Use machine learning to predict patient flow and automatically optimize appointment slots.
- **Mobile Application Development:**
Provide a mobile app for easy booking, tracking, and notifications.
- **Real-Time Notifications:**
Enable SMS/email alerts for appointment confirmation, reminders, and updates.
- **Cloud Integration:**
Deploy the system on cloud platforms for better scalability, security, and remote access.
- **Multi-Hospital Support:**
Extend the system to manage appointments across multiple hospitals or branches.
- **Advanced Patient Prioritization:**
Use medical history and condition severity for more accurate priority handling.

Conclusion

The Hospital Appointment Scheduling System provides an efficient solution for managing patient appointments by automating the scheduling process. It reduces waiting time, minimizes conflicts, and ensures better utilization of doctors through priority-based scheduling.

By using optimization techniques such as Greedy Algorithm, Genetic Algorithm, and Simulated Annealing, the system generates effective and reliable schedules. It also improves patient experience and reduces manual effort.

Overall, the system demonstrates how intelligent scheduling can enhance hospital management and can be further extended for real-world healthcare applications.

The background features a series of overlapping, curved shapes in shades of blue and white, creating a dynamic, flowing effect. The shapes are smooth and organic, resembling stylized waves or abstract architectural elements. The colors range from a deep navy blue to a bright, vibrant blue, with white filling the spaces between the curves.

THANK YOU